Complex Analysis Homework 5

1. (a) Suppose $p(z) = a_n z^n + \cdots + a_1 z + a_0$ is bounded by M for $|z| = 1$. Show that each coefficient a_k is bounded by M.

(b) Suppose $p(z) = a_n z^n + \cdots + a_1 z + a_0$ is bounded by 1 for $|z| \le 1$. Show that $|p(z)| \le |z|^n$ for all z where $|z| \ge 1$.

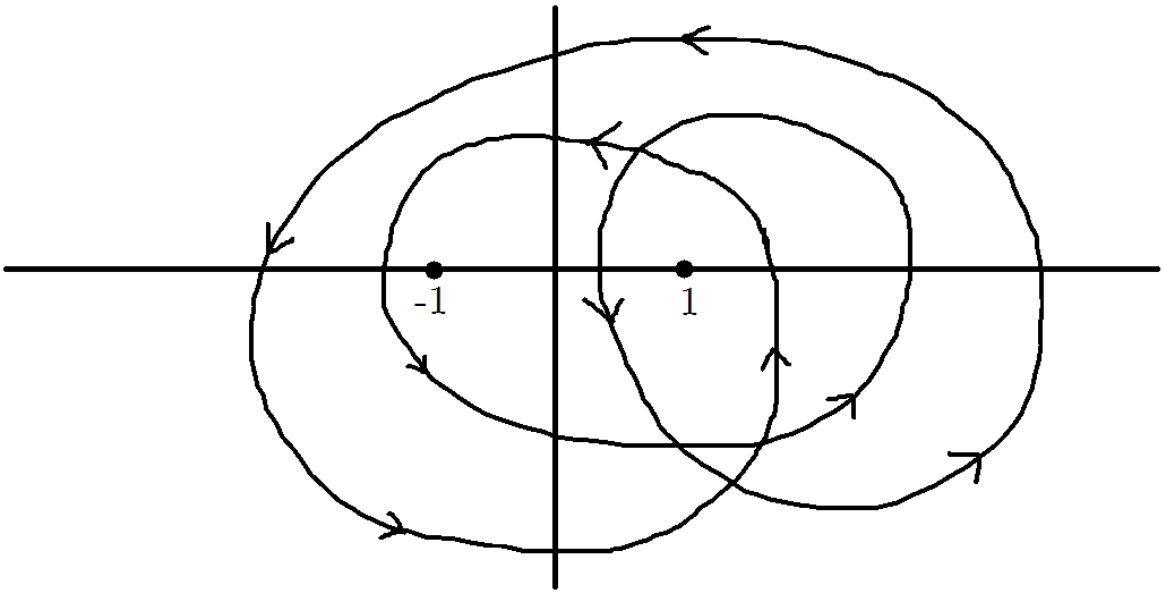
Hint: Consider $q(z) = \dfrac{p(z)}{z^n}$ in the annulus: $1 \le |z| \le R$ for "large" R.]

2. Let $P(z) = z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0, \quad n \ge 2,$ such that all its zeros $c_1, c_2, \cdots, c_n$ lie inside the circle $|z| = R$.

(a) Show that $\left| \dfrac{1}{P(z)} \right| \le \dfrac{1}{(R - |c_1|)(R - |c_2|) \cdots (R - |c_n|)}$

(b) Hence show that $\oint_{|z|=R} \dfrac{1}{P(z)} dz = 0.$

3. Let C be the curve in below figure. Calculate $\int_C \frac{e^z}{z^3 - z^2 - z + 1} dz$.



4. If $f(z)$ is analytic on and inside the unit circle $|z| = 1$, show that

$$(1-|z|^2) f(z) = \frac{1}{2\pi i} \oint_{|\zeta|=1} f(\zeta) \frac{1-\bar{z}\zeta}{\zeta - z} d\zeta, \quad |z| < 1,$$

and deduce that

$$(1-|z|^2)|f(z)| \le \frac{1}{2\pi} \int_0^{2\pi} |f(e^{i\theta})| d\theta, \quad |z| < 1.$$

5. Evaluate the following integrals with counterclockwise contour:

(a) $\oint_{|z|=2} \frac{z^n}{z-1} dz, \quad n \ge 0$

(b) $\oint_{|z|=1} \frac{z^n}{z-2} dz, \quad n \ge 0$

(c) $\oint_{|z|=1} \frac{\sin z}{z} dz$

(d) $\displaystyle\int_{|z-1-i|=5/4} \frac{\operatorname{Log} z}{(z-1)^2}\,dz$

(e) $\displaystyle\int_{|z|=1} \frac{dz}{z^2(z^2-4)e^z}$

(f) $\displaystyle\int_{|z-1|=2} \frac{dz}{z(z^2-4)e^z}$